Grounded Enterprise Policy & Procedure Assistant

1. Problem and Business Value

Procurement and finance staff often need to determine whether a purchase is permitted, what approval is required, and which supplier or documentation rules apply. In practice, relevant policies may be distributed across multiple documents, contain numerical thresholds and conditional rules, and include outdated or conflicting provisions. Conventional keyword search can locate relevant passages, but it does not determine whether the evidence is sufficient, clarify missing information, or explain which rule should be applied.

I therefore built a Grounded Enterprise Policy & Procedure Assistant as a decision-support system rather than an autonomous procurement system. The system uses 16 policy documents, including 15 current documents and one explicitly superseded policy edition. The corpus also contains a genuine cross-document contradiction involving gift acceptance. The target user is a procurement or finance staff member who needs a fast, evidence-based policy answer.

The project's intended improvement over the current workflow is not simply faster search. It is to transform scattered policy text into a structured interaction: retrieve evidence, determine whether the question is sufficiently specified, answer only when the evidence supports an answer, and otherwise clarify or abstain.

2. Technical Approach and Key Trade-offs

The final architecture combines deterministic retrieval and decision logic with a Foundation Model. A user question is first processed using TF-IDF paragraph retrieval with numerical and domain-specific ranking bonuses. The top five passages are then passed to a deterministic decision layer. This layer can produce three outcomes: CLARIFY, ABSTAIN, or ANSWER. Only answer-eligible cases are sent to openai/gpt-4o-mini through OpenRouter, and the model is instructed to generate an answer only from the retrieved evidence and include citations.

A major technical trade-off was using local TF-IDF retrieval instead of a more sophisticated hosted vector-search or embedding service. Local retrieval provides low cost, reproducibility, transparent behaviour, and no additional infrastructure. However, it is more vulnerable to vocabulary mismatch. Earlier retrieval stress testing showed that semantically related questions can sometimes give slightly higher scores to lexically different documents. Numerical and domain signals were therefore added to improve ranking without turning retrieval scores into probabilities.

Another important trade-off was deterministic decision-making versus allowing the Foundation Model to decide whether a question can be answered. I chose deterministic rules because clarification and abstention are safety-critical behaviours. Thresholds were calibrated separately

on 90 questions and frozen before the final 22-case evaluation. This makes the routing decision reproducible and prevents the model from silently deciding that weak evidence is sufficient.

The Foundation Model is therefore used for bounded synthesis rather than as the policy authority. This introduces API cost and latency, but allows the system to convert multiple retrieved passages into a readable, cited response. It also creates a remaining failure mode: correct retrieval does not guarantee that the generated answer discloses every important qualification.

3. Evaluation Results and Performance Tuning

The final evaluation used 22 frozen cases: five normal cases, five boundary cases, five ambiguous cases, five insufficient-evidence cases, and two conflicting/outdated cases. Calibration data were kept separate from the final test set.

The primary metric was citation-grounded answer correctness for answerable cases. The system achieved 10/12 (83.33%) on this metric. Deterministic action correctness was 22/22 (100%), evidence grounding was 17/17 (100%), ambiguous-question clarification was 5/5, and insufficient-evidence abstention was 5/5. The two conflict/outdated cases also achieved the correct action-level handling.

The baseline comparison provides context. Both the majority-class and local keyword-search baselines achieved 54.5% decision accuracy on the frozen label distribution. The keyword-search baseline can retrieve relevant passages but does not have an explicit mechanism for clarification or abstention. The proposed system therefore adds value primarily through controlled decision routing and grounded synthesis, rather than simply replacing keyword search with a more complex retrieval model.

Performance tuning focused on retrieval thresholds and routing behaviour rather than tuning against the final test set. The final thresholds were frozen after calibration. The system also includes structured evidence overrides and a multi-source rule for cases where evidence from more than one policy document is necessary.

The final system made 15 Foundation Model calls across the 22 test cases. Total API cost was approximately US$0.004831. Median model latency was approximately 2.09 seconds, with a P95 of approximately 5.77 seconds. End-to-end median latency was approximately 1.55 seconds.

4. Difficulties and Evaluation Critique

The most important difficulty was not retrieval itself but ensuring that successful retrieval translated into a complete answer. Two final cases demonstrate this limitation.

In TC21, the system retrieved the relevant Ontario policy evidence but failed to explicitly disclose that the cited 2019 policy was superseded. In TC22, the system returned a bare negative answer

instead of clearly disclosing the unresolved conflict between the gift-acceptance provisions in two policy documents and recommending escalation.

These failures are important because they show that grounding is not equivalent to complete disclosure. The evidence can be correct while the final response still omits a material qualification. This is why the evaluation separates evidence grounding from citation-grounded answer correctness.

The evaluation itself also has limitations. The corpus is synthetic and relatively small, and the policies were written within a controlled project context. The 22-case frozen test set provides structured coverage of important behaviours but cannot represent the full variability of real procurement questions. In addition, human answer grading introduces judgement, although the final answer texts were manually checked against the expected evidence and ground truth.

5. Business and Technical Limitations

The system intentionally does not approve purchases, place orders, execute transactions, or make autonomous procurement decisions. Human verification remains necessary for consequential decisions. This reduces the degree of automation but provides a clearer accountability boundary.

The current TF-IDF approach is also appropriate mainly for a small, controlled corpus. A larger enterprise policy repository would likely require stronger semantic retrieval, document version management, access control, and monitoring. The current API dependency also introduces external service availability, cost, and data-governance considerations.

A separate user study was designed to measure task-completion time against manual search, but no participant data had been collected at the time of submission. Therefore, the previously proposed 50% reduction target is reported only as a future target and not as a measured project result.

6. Future Path and Conclusion

The next stage would focus on improving disclosure rather than simply increasing model complexity. Supersession and contradiction detection could become explicit structured signals in the decision layer, while semantic retrieval could complement TF-IDF for larger corpora. Independent users should then evaluate both correctness and task-completion time on realistic enterprise policy questions.

Overall, the project demonstrates a practical trade-off between automation and control. The final system achieved 22/22 deterministic routing correctness and 17/17 evidence grounding, while citation-grounded answer correctness reached 10/12. Its remaining failures reveal an important boundary: a grounded AI system can retrieve the right evidence while still communicating an incomplete conclusion. The resulting system is therefore best positioned as a grounded enterprise

decision-support assistant with human verification, rather than an autonomous procurement decision-maker.